

The empirical recurrence for OEIS A235928: recovering a conjecture that is not written down

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Abstract

OEIS A235928 records “Empirical recurrence of order 70”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 70$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 70, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 70 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235928 is “Number of $(n+1) \times (3+1)$ 0..3 arrays with the minimum plus the maximum unequal to the lower median plus the upper median in every 2×2 subblock.”. It has offset 1 and begins

18992, 1490648, 116219056, 9098402456, 710992703856, 55608647673784, 4347481602940704, 33995548526908534

The entry states:

Empirical recurrence of order 70 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:56:09, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 70$ of the 256, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 70$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 512$ exact terms of the model returns order 70 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{70} c_i a(n-i),$$

c_1	0	c_{19}	1747934095273082842708380	c_{37}	−6457202704793849295
c_2	6796	c_{20}	−62500563894669356384908659	c_{38}	−75277315148021380284
c_3	113468	c_{21}	−114166856431414805417571904	c_{39}	−62757259865611135869
c_4	−12185323	c_{22}	2883913926832371798189453666	c_{40}	249925767736467376676
c_5	−225122106	c_{23}	6654246584966097400086520650	c_{41}	511295002458401537558
c_6	11127198829	c_{24}	−99179498172089037462425050696	c_{42}	−24973665976557416156
c_7	184814714126	c_{25}	−307003400962368155814948046488	c_{43}	−158139623638928702002
c_8	−6294151148133	c_{26}	2472206598757043641402081652706	c_{44}	−77102900154621678284
c_9	−81607265345366	c_{27}	10515767800646627241213294586532	c_{45}	2421711278767419370393
c_{10}	2363983111320163	c_{28}	−41422833502739409451238095805104	c_{46}	2839177012318787218738
c_{11}	21152235213336922	c_{29}	−258766487369942228970078796807402	c_{47}	−166781023971719071491
c_{12}	−598507563900684936	c_{30}	360582235585167395161318737159774	c_{48}	−412409266370509665244
c_{13}	−3352148231749121024	c_{31}	4427904377562676483146584462195460	c_{49}	−13807744076974397410
c_{14}	102436960717527436426	c_{32}	1488667596810776074735933299069012	c_{50}	3367441596171608059167
c_{15}	344030365409220305754	c_{33}	−49836902318783146405077485885132592	c_{51}	1124107479072148789377
c_{16}	−12062703487489176036408	c_{34}	−87012643800858641701720229569283888	c_{52}	−169273514997059681972
c_{17}	−26055137419813860316022	c_{35}	324005716150290674214956791618553824	c_{53}	−89208724015670984420
c_{18}	1012506131970720361585140	c_{36}	1136520082951907337176204426020774336	c_{54}	550692681566905466697

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 70, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $70 + 4$ terms removed returns order 70 as well, so $\deg q_{\min} = 70 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 11 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $70 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 70 that this sequence satisfies — and that family turns out to have exactly one member.

References

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